

Abraham Khan

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PhD Candidate in Mathematics

Overview

As a computational mathematician, I develop fast algorithms for densely structured linear and multilinear algebra problems. My current research focuses on developing tensor-based algorithms with an eye towards applications in scientific computing and machine learning; in particular, my focus is on efficiently computing kernel matrices. My broader vision is to develop efficient numerical methods that scale efficiently for larger problem sizes and mitigate the curse of dimensionality: the ultimate goal is for the complexity to grow linearly with the problem's intrinsic dimension and overall size.

Research Interests

Numerical linear algebra; multilinear algebra; machine learning; approximation theory; theoretical computer science.

Education

Ph.D., Mathematics, North Carolina State University

2020–Expected March 2026

Raleigh, NC

Dissertation: Parametric kernel approximations using tensors.

Advisor: Dr. Arvind K. Saibaba.

Qualifying exams: Matrix Theory; Functional Analysis; Abstract Algebra.

B.S., Mathematics, University of South Carolina

2014–2018

Columbia, SC

B.S., Computer Science, University of South Carolina

2014–2018

Columbia, SC

Research Experience & Appointments

Research Assistant, North Carolina State University

2022–Present

Raleigh, NC

Advisor: Dr. Arvind K. Saibaba

NSF RTG Trainee in Randomized Numerical Analysis

Work supported by NSF grants (DMS-1745654, DMS-1845406, DMS-2411198)

- Developed the *Parametric Tensor Train Kernel (PTTK)* method, a tensor-train-based method for computing parametric low-rank approximations of parametric matrices across the full parameter domain.
- Developed *parametric hierarchical matrices*, a new class of hierarchical matrices that efficiently approximates parametric kernel matrices across the full parameter domain.
- Currently investigating hierarchical matrix approximation using tensor decompositions.

Givens Intern, Argonne National Laboratory

Summer 2024

Lemont, IL

Mentor: Dr. Vishwas Rao

- Explored data-sparse approximations (hierarchical matrices) and developed the concept of parametric hierarchical matrices.
- Continued this work after the internship to further develop the parametric hierarchical matrix framework.

Undergraduate Researcher, University of South Carolina

2016–2019

Columbia, SC

Mentor: Dr. Joshua Cooper

- Explored a problem that is at the intersection of order theory and theoretical computer science: determine the computational complexity of counting the number of linear extensions of a dimension two semiorder.

Publications

- **Khan, A.;** Saibaba, A. K. (2025). *Parametric kernel low-rank approximations using tensor train decomposition*. *SIAM Journal on Matrix Analysis and Applications*, **46**(2), 1006–1036. (33 pages).
- **Khan, A.;** Chen, C.; Rao, V.; Saibaba, A. K. (2025). *Parametric hierarchical matrix approximations to kernel matrices*. arXiv preprint [arXiv:2511.03109](https://arxiv.org/abs/2511.03109) [math.NA]. (39 pages).
- Scott, M.; **Khan, A.;** Kilmer, M. E.; Hu, X.; Saibaba, A. K. *Hierarchical matrix approximation using tensor decompositions*. Manuscript in preparation.

Software

- **PTTK Method** — <https://github.com/awkhan3/ParametricTensorTrainKernel>. Implementation of the PTTK method (Python).
- **Parametric Hierarchical Matrices** — <https://github.com/awkhan3/ParametricHierarchicalMatrices>. Implementation of the parametric hierarchical matrix approach (Python).
- **TensorTrain Greedy-Cross** — <https://github.com/awkhan3/TensorTrain-Greedy-Cross>. Implementation of the greedy restricted cross interpolation algorithm for tensor trains (Python).

Conferences & Presentations

SIAM Conference on Computational Science and Engineering (CSE25) *March 3–7, 2025*

Fort Worth, TX

Talk: “Parametric Kernel Low-Rank Approximations Using Tensor Train Decomposition.”

SIAM Conference on Mathematics of Data Science (MDS24)

October 21–25, 2024

Atlanta, GA

Poster: “Parametric Kernel Low-Rank Approximations Using Tensor Train.”

Workshop on Sparse Tensor Computations

October 18–19, 2023

Chicago, IL

Poster: “Efficient Low-Rank Parametric Kernel Matrix Approximation Using the Tensor Train Decomposition.”

Internal Talks

Randomized Numerical Analysis RTG

November 8, 2024

Raleigh, NC

Talk: “Parametric Kernel Low-Rank Approximations Using Tensor Train.”

Computing Skills

Python · C++ · C · MATLAB · Git · Linux · HPC

Teaching Experience

Teaching Assistant, MA 305: Introductory Linear Algebra and Matrices, North Carolina State University

Fall 2020

Raleigh, NC

- Held office hours; graded assignments and exams.

Teaching Assistant, MA 242: Calculus III, North Carolina State University

2021–2022

Raleigh, NC

Terms: Spring 2021; Fall 2021; Fall 2022.

- Led recitations; held office hours; graded assignments and exams.

Teaching Assistant, MA 121: Elements of Calculus, North Carolina State University

Spring 2022

Raleigh, NC

- Held office hours; graded assignments and exams.